

Need: First year EngSci students need a way to rest in campus buildings in between classes

Goals:

- Private
 - Decrease sound by x decibels
 - The amount of light that can go through is x lumens?
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- Portable
 - Mass is less than x kg
 - Volume is less than $x \text{ m}^3$
 - Can be taken out in x seconds/minutes
 - Can be put back in x seconds/minutes
- Comfortable
 - Maximum pressure at any point on human body is less than x Pa
 - Conforms to human body shape, reference anthropomorphic data
- Should have a way to wake up user before class
 - Decibel of alarm
 - Failure rate of alarm %?
- Protected from water, dust, etc.
 - IP standards

Reference Designs:

- Inflatable pillows
- Clip on headrest
- Small tents
- Nap pods
- Blanket jackets

Need: First Year EngSci students need a reliable umbrella to stay dry when traveling to class when it is rainy and windy.

Goals:

- It should not flip inside out when it is windy
 - Withstand x speed winds
- Should still be portable and easy to carry
 - Objective: dimensions
 - Objective: cannot have multiple pieces
- Should be deployable and usable by all users, no matter muscle strength and height.
 - Objective: mass
 - Objective: when it catches wind, a user with general muscle mass should be able to hold the umbrella still
 - Usually if you're traveling against the wind, and angle the umbrella so that the rain doesn't hit your face, the wind is too strong, and pushes back on the umbrella due to its shape. This wind pushing back on the umbrella makes it harder to hold in place.
- Should not store water, it must not make the surroundings it is put onto wet after use

Objectives:

Reference Designs:

- Inverted Umbrellas
 - Issue: I have this umbrella, and from personal experience, it don't work, the classroom tables still get wet
- Carbon fibre umbrellas -> expensive EngSci's aint got that much money

Need: To store papers efficiently and easy access

Goals:

- Papers should not fall out
 - Be able to store an x amount of papers without the binder rings popping open
- Papers should not tear
 - Paper should not experience more than 1 kg of force where the rings are in contact with the paper
 - Paper should not get caught in the zipper of the binder
- Environmental protection of paper
 - Sealed binder with an ip rating of ipx6
- Easy to insert and extract papers
 - Have a higher minimum force to open the binder ring to prevent accidental "pops"

Reference Designs:

- Clicking button to open and close the binder ring instead of applying force on the rings themselves
- Use of waterproof material on the outside of the binder and sealed off by a zipper
- Instead of round rings, use a flat edge ring to be in contact with the paper to increase the contact surface area and distribute the force experienced by the paper over a larger area
- Set a clear point for which the user can know that their binder is now full, max line or something like that

Need: Portable chargers are often slow to recharge and run out quickly → inefficiency and inconvenient.

Goals:

- Solar-powered charger with foldable panels
- Constant automatic charging
- Sufficient charging speed for smartphones
- Solar panel efficiency: X amount of recharge per minute
- providing power without ever dying out. No manual recharge needed
- Fast and sustainable energy
- Portable → under x kg
- Easy storage → x dimensions
- Wireless for easy use

Reference Designs

- Small wireless charging that sticks to back of phone with same dimensions with phone